

(Note: Complete all the question in your Machine Learning Notebook)

Baye's Theorem Numerical

1. Imagine there are 1000 people in a town, and 10 of them have a particular disease. There is a test for the disease that correctly identifies the disease in 9 out of 10 people who have it. However, it also incorrectly identifies the disease in 1 out of 100 people who do not have it. If a person tests positive, what is the probability that they actually have the disease?

- Hint: Use Bayes' Theorem with the following:
 - Number of people with the disease: 10
 - Number of people without the disease: 990
 - Number of true positives: 9
 - Number of false positives: 10

2. Out of 2000 people in a city, 10 have a certain disease. A test is available that correctly identifies 8 out of 10 people who have the disease and incorrectly identifies 1 out of 50 people who do not have the disease. If someone tests positive, what is the probability that they actually have the disease?

- Hint: Use Bayes' Theorem with the following:
 - Number of people with the disease: 10
 - Number of people without the disease: 1990
 - Number of true positives: 8
 - Number of false positives: 40

3. Imagine you receive 100 emails, and 20 of them are spam. The word "free" appears in 10 of the spam emails and in 4 of the non-spam emails. If an email contains the word "free," what is the probability that it is spam?

- Hint: Use Bayes' Theorem with the following:
 - Number of spam emails: 20
 - Number of non-spam emails: 80
 - Number of "free" in spam: 10
 - Number of "free" in non-spam: 4

4. In a group of 500 people, 20 have a particular disease. A test for the disease correctly identifies 18 out of 20 people who have the disease and incorrectly identifies 5 out of 480 people who do not have the disease. If a person tests positive, what is the probability that they actually have the disease?

- Hint: Use Bayes' Theorem with the following:
 - Number of people with the disease: 20
 - Number of people without the disease: 480
 - Number of true positives: 18
 - Number of false positives: 5

5. You have two coins. One is a fair coin, and the other lands heads 3 out of 4 times. You pick one coin at random and flip it, getting heads. What is the probability that you picked the biased coin?

- Hint: Use Bayes' Theorem with the following:
 - Number of fair coins: 1
 - Number of biased coins: 1
 - Number of heads with the fair coin: 1 out of 2
 - Number of heads with the biased coin: 3 out of 4